ML Course Project

Project Title = Multi-task Fine-tuning of Flan-T5 for Code Intelligence

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Project Description

Large language models are increasingly used for **developer productivity tasks** such as generating code, translating natural language to SQL, and explaining code. While instruction-tuned models like FLAN-T5 already demonstrate strong zero-shot capabilities, adapting them efficiently to **multiple programming-related tasks** remains an open challenge.

This project explores how fine-tuning flan t5 using LoRA can improve performance on **code intelligence tasks**, including:

- Natural Language → **Python code generation**
- Natural Language → **SQL query generation**